

Overview and Objectives

In this course, we will cover topics dealing with various programming languages, comparing their programming paradigms, syntax, and purposes. We will also talk about project management, software engineering, and compilers.

Different programming languages will be used in the projects to demonstrate syntax comparisons between various languages. Python, Java, LISP, Prolog, and Go will be covered. We will become familiar with the properties of languages and common metrics by which they can be compared.

We will also introduce some common ideas of software engineering, project management, and best practices which will be used to complete various these projects throughout the semester. These topics will include using version control systems, creating and maintaining software design documents, and following an iterative design process.

Essential Information

Instructor:	Erik Steinmetz
E-mail:	steinmee@augsborg.edu
Phone:	(612) 330-1062
Office:	Hagfors 369E
Office Hours:	MW 1:30 pm - 2:00 pm, T 10:30 am - 11:00 am, or by appointment
Class Location:	Hagfors 367 Advanced Lab
Class Times:	TTh 8:30 to 10:00 am
Class Dates:	20 January through 30 April, 2026
Final Quiz:	Tuesday 6 May, 2025 at 8:30 am in Hagfors 367
Textbooks:	“Concepts of Programming Languages” (12th Edition); Sebesta eText at: https://console.pearson.com/enrollment/8rujs1 “Foundations of Computing”; Allison
Prerequisites:	Completion of MAT 202/302 and CSC 371 with a C- or better.

Grading

Grades will be determined based on the accumulation of up to 1000 points. There will be five main projects during the semester, each using a different programming language and accounting for 80 points (total of 40% of your grade). There will be eleven quizzes at 40 point each, dropping the lowest quiz (total of 40% of your grade). The remaining 20% will be given for various assignments and class activities. A few of the assignments involve writing code, so you should expect to be writing some code every day in order to complete your projects and assignments.

In order to earn full points for any given project, your work will need to:

- Work correctly and be tested.
- Be properly commented and documented.
- Adhere to proper naming standards and conventions for the language.
- Be correctly organized and updated in a git repository.

Each project is submitted in three stages: design, first iteration, and final iteration. Late project stages will be accepted up to one week late, but points awarded for that stage will be discounted by 50%. Late assignments or quizzes will not be accepted. Turn your work in before the deadline and make sure that it is visible on Gitlab or Moodle or both.

The entire one thousand points are tallied at the end of the semester. Typically $total \geq 900$ will be various flavors of A grades, $total \geq 800$ B grades, and $total \geq 700$ C grades. $700 > total > 550$ will result in a D grade, while lower will be an F. Earning less than 70% of the quiz points ($quiztotal \leq 280$) will result in a D or F grade.

Please note that a C- or higher is required for a pass under the pass / no-pass grading option and is required for the course to apply to the Computer Science major or minor.

Expectations

- **Attend every class session.** Most of our class sessions will be some combination of lecture, working through examples of writing code, activities, and project help. It is important that you make use of this time to get your questions answered. You should bring your laptop during lab sessions, or borrow one from Lindell library.
- **Use class time effectively and allow others to do the same.** Our class time is for discussing, reading, learning, and writing about programming languages and how they are put together. It is not time for surfing social networking sites, watching cat videos, or texting friends. No use of phones is allowed in class other than taking photos of the whiteboard. No use of laptops other than writing code or for approved accommodations is permitted.
- **Submit assignments and code when due.** A substantial part of this course is the writing of a large amount of code. In order to remain current, please turn in your code on time.
- **Write your own code.** All the work you submit, whether code or other assignments, must be your own work. You are encouraged to discuss ideas, solutions, explanations, and references with your classmates. You are also encouraged to work together as you do assignments and write code, but everything you submit must be your own handiwork. Please refer to the Augsburg Academic Honesty Policy which can be found at <https://tinyurl.com/AcademicHonestyAugsburg>. Please note that penalties for breaching academic honesty policies include receiving a zero on the assignment and may result in receiving an F in the course. To establish a clear division between acceptable and unacceptable sharing, no electronic copying of code is allowed: you must type everything in yourself. Do not send or receive code from your classmates and do not copy-paste code from any other source. Both sender and receiver of copied materials are considered in violation of the no-copying rule.
- **Schedule your time.** Reading materials from the text, working on assignments, and writing the substantial amount of code for this course is time consuming. Expect to spend ten hours or more each week reading, writing, and coding outside of class time to succeed in this class.
- **Pay attention to Moodle and email.** Our schedule and assignments are posted on Moodle and may be updated there. I will send email notifying you of any significant changes or updates.

Schedule of Topics

We will be covering a substantial portion of the Sebesta “Concepts of Programming Languages” (12th Edition) textbook, but not entirely in textbook order. Here are the parts we will cover.

- Introduction - Chapter 1
- History - Chapter 2
- Lexical Analysis - Chapter 4 (sections 1 and 2)
- Syntax and Semantics - Chapter 3 (sections 1 through 3)
- Names and Scope - Chapter 5
- Data Types - Chapter 6
- Expressions and Assignment - Chapter 7
- Functional Programming Languages - Chapter 15
- Logic Programming Languages - Chapter 16
- Control Structures - Chapter 8
- Subprograms - Chapter 9
- Implementing Subprograms - Chapter 10
- Concurrency - Chapter 13

In parallel to covering the material in the textbook we will be coding five programming projects in five different programming languages.

- **Python** - Procedural Programming
- **Java** - Object-Oriented Programming
- **LISP** - Functional Programming
- **Prolog** - Logic Programming
- **Go** - Concurrent Programming

Additional Information

Accessing Online Resources

Information about getting a computer or an internet connection from the University can be found at <https://sites.augsburg.edu/it/services-we-provide/student-access-to-computing-resources/>.

Disability Accommodations and Accessibility

Students with disabilities who believe that they may need accommodations in this class are encouraged to contact the Center for Learning and Adaptive Student Services (CLASS) (612-330-1053 or stop by the Gage Center welcome desk) as soon as possible to better ensure that such accommodations are implemented in a timely fashion. Contact info about the CLASS program can be found at <http://www.augsburg.edu/class/>. Be aware that it is your responsibility to meet the requirements of your CLASS office accommodations.

Title IX

Augsburg University is committed to creating a safe learning environment for all students, including one that is free of sexual misconduct. If you or someone you know has experienced sexual misconduct, including sexual assault, dating violence, domestic violence, and stalking, you may contact Michael Grewe, Ph.D., LISW, Assistant Provost & Title IX Coordinator, at titleix@augsb.org or 612-330-1499 to report an incident, seek support, and/or take action.

Please be aware that faculty members are required to disclose information about suspected or alleged sexual misconduct or other potential violations of the Augsburg University Sexual Misconduct Policy to the Title IX Coordinator. If the Title IX Coordinator receives information about an incident, they will reach out to offer information about resources, rights, and procedural options as a member of the campus community. If you or another student you know wishes to speak to a confidential resource who does not have this reporting responsibility, you may contact the Center for Wellness and Counseling at cwc@augsb.org or 612-330-1707; Campus Ministry at 612-330-1732; or the Aurora Center at aurora@umn.edu or 612-626-9111 (24/7).

FERPA

As your instructor, I am committed to protecting your privacy by only using university-approved course technologies and adhering to the Family Educational Rights and Privacy Act (FERPA). This includes using your educational data only as allowed by FERPA. I am unable to share any personally identifiable information without your written consent, except for legitimate educational purposes, or under certain conditions described above. FERPA applies to all students enrolled in this institution regardless of sex, race, color, national or ethnic origin, religious belief, age, disability, gender, sexual orientation, gender identity or expression, marital status, familial status, pregnancy, citizenship, creed, genetic information, or veteran status.

Disclaimer

Policies stated within this syllabus are subject to change, following verbal announcement in class. Students are responsible for knowing the current version, always to be found on the course Moodle page.